Recall: Operations:

- Complement: $\overline{R} = \{(a,b) \in A \times B \mid (a,b) \notin R\}$
- Inverse: $R^{-1} = \{(b,a) \mid (a,b) \in R\}$ relation from B to A
- Composition: If $R \subseteq A \times B$, $S \subseteq B \times C$, then

  $S \circ R = \{(a,c) \in A \times C \mid \text{there exists } b \in B \text{ s.t. } (a,b) \in R, (b,c) \in S\}$
  relation from A to C

Class activity:

$A = \{1, 2, 3\}$, $B = \{(1,2,3,4)\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (3,4)\}$

find

$\overline{R}$, R^{-1}, $R \circ R^{-1}$, and $R^{-1} \circ R$

§9.3: Representing relations

Option 1: Use a matrix!

$A = \{a_1, a_2, \ldots, a_m\}$

$B = \{b_1, \ldots, b_n\}$

$R \subseteq A \times B$

relation

Then R can be represented by the matrix

$$M_R = [m_{ij}] \text{ where } m_{ij} = \begin{cases} 1, \text{ if } (a_i, b_j) \in R \\ 0, \text{ otherwise} \end{cases}$$

Ex 1: $A = \{1,2,3\}$ $B = \{1,2\}$

$R = \{(2,1), (3,1), (3,2)\}$

$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Class activity (Ex 3):

Let R be a relation on A s.t.

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Find R. Is it reflexive? Symmetric? Antisymmetric?

If R & S are rel'ns on A, then

$$M_{R \cup S} = M_R \cup M_S$$

an entry is 1 if the corresp. entry in <u>either</u> M_R or M_S is 1

$$M_{R \cap S} = M_R \cap M_S$$

same, but replace "either" with "<u>both</u>"

$$M_{S \circ R} = M_R \odot M_S$$

boolean prod: take matrix product, then change any pos. num. to 1

<u>note</u>: order reversed!

Ex: If $M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$M_{R \cup S} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $\quad$ $M_{R \cup S} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$M_R M_S = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 3 & 1 \\ 1 & 2 & 1 \end{bmatrix}$,

so $M_{S \circ R} = M_R \odot M_S = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Option 2: Use a graph!

not the x,y kind of graph

Def: A <u>directed graph</u> or <u>digraph</u> consists of a set V of <u>vertices</u> and a set E of <u>edges</u>. Each edge goes from one vertex to another

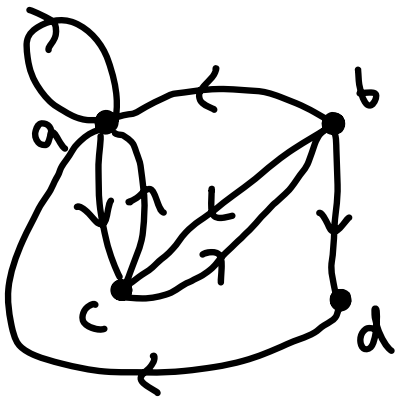
Let R be a relation on A. To make a digraph for R,

V: vertices labelled by elts. of A

E: edge going from a to b if $(a,b) \in R$

Ex 8: $A = \{a,b,c,d\}$

$R = \{(a,a),(a,c),(b,a),(b,c),(b,d),(c,a),(c,b),(d,a)\}$



Reflexive: loop at every vertex

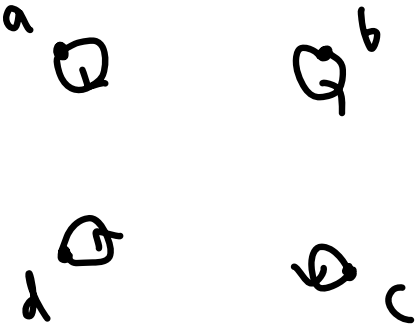
Symmetric: edges always come in opposite-direction pairs

Antisymmetric: edges never come in opposite-direction pairs

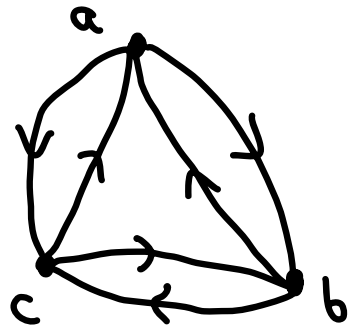
Transitive: For every length-2 path, there is a single edge w/ the same start and end

Class activity: Are the following rel'ns reflexive/symmetric/antisymmetric/transitive? Draw the corresponding matrix/digraph.

a)



b)



c)

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$